Rows Columns and Data Types

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Rows Columns and Data Types before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Debugging Strategies

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Trace Tables

Without opening the old lesson, explain one key idea from Trace Tables. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Precision in Algorithms

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Rows Columns and Data Types trace and record the first step you can explain without help.

Use W2: Classify records for Rows **column**s and **schema**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Rows Columns and Data Types explanation against the supplied facts before discussing it.

Use the worked case for Rows **column**s and **schema**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Design a schema independently

For a fictional shelf audit, define label and book_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Rows Columns and Data Types target to a trace and a justified conclusion.

A row contains title='Sky', pages='120', available='false'. Explain why treating nonempty text as true can misread availability. Propose explicit conversions and a rejected-**input** case.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Rows Columns and Data Types case and show the deciding evidence.

Fields are Title, Pages and Borrowed; a row contains River, ninety and yes.

Assign intended **type**s and a repair.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A table row is book_id B7, pages 84, available true. Name the **type** of each field and explain why pages stored as eighty-four complicates calculation.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 7 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Rows Columns and Data Types adds to Transparent Book Recommender and keep the evidence.

Store one book per row and keep reader preferences in a separate record.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A row contains title='Sky', pages='120', available='false'. Explain why treating nonempty text as true can misread availability. Propose explicit conversions and a rejected-**input** case.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 State a schema

> State a schema. Apply the supplied Rows Columns and Data Types facts and keep your reasoning.

List the required fields, **type**s and duration boundaries for the worked table.

> **My reasoning and evidence**

WORKBOOK · 13 OF 21

W2 Classify records

> Classify records. Apply the supplied Rows Columns and Data Types facts and keep your reasoning.

Classify minutes values 0, 60, -1 and 61 under this **schema**.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Repair a unit problem

> Repair a unit problem. Apply the supplied Rows Columns and Data Types facts and keep your reasoning.

A source says 180 seconds but the row says minutes: 180. Give a documented correction.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Design a schema independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Design a schema independently. Apply the supplied Rows Columns and Data Types facts and keep your reasoning.

For a fictional shelf audit, define label and book_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Separate validity and truth

Separate validity and truth. Apply the supplied Rows Columns and Data Types facts and keep your reasoning.

A row says Reading, 12 and passes. Does that establish the session really lasted **12 minutes**?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Rows Columns and Data Types

> A fictional approved book catalogue has three fields per row. Explain the labelled step before checking the changed condition.

A fictional approved book catalogue has three fields per row.

Schema: id is nonempty text; pages is an integer 1–300; illustrated is Boolean True or False.

Rows A={id:"B7",pages:90,illustrated:True}; B={id:"B8",pages:"90",illustrated:False}; C={id:"",pages:80,illustrated:True}.

AI ACADEMY / AI-2/7

Rows Columns and Data Types

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Column contract

Field names identify values; order is not the only guide.

↓

2. Validate A

B7 is text, 90 is allowed integer, True is Boolean: valid.

↓

3. Validate B

Quoted "90" is text, so pages fails the integer contract.

↓

4. Validate C

Empty id fails even though its other two values have allowed types.

CHANGE ONE THING

Change B's pages from "90" to integer 90 using an approved repair.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which two rows originally fail, and do they fail for the same reason?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Publish the recommender's row **schema** and validation results before ranking books, preserving repair provenance.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Publish the recommender's row **schema** and validation results before ranking books, preserving repair provenance.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional approved book catalogue has three fields per row.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Empty id fails even though its other two values have allowed **type**s.

1. **Column** contract

2. Validate C

3. Validate B

4. Validate A

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Validate C Empty id fails even though its other two values have allowed **type**s. A **schema** makes row meanings and expected **type**s explicit.

The string "90" looks like a number when printed but does not meet this strict integer **schema**. A repair can be authorised later, but must not be assumed silently.

Missing or empty identifiers also prevent tracing an item back to its source.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change B's pages from "90" to integer 90 using an approved repair. Which conclusion is supported by the supplied facts?

1. B now satisfies the declared **schema**.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: B now satisfies the declared

schema. Its id is nonempty and illustrated is Boolean; the page-**type** defect is removed.

Repair a missing building block

Review Debugging Strategies (ai-2/6). Goal: Design revealing tests for a function. Short practice: Which test most directly isolates the handling of loose books, and what should it produce?.

Return to this week and explain what you changed.

Review Trace Tables (ai-2/4). Goal: Trace variable values before and after each loop iteration. Short practice: Move print inside the original loop. What sequence is displayed?.

Return to this week and explain what you changed.

Review Precision in Algorithms (ai-2/1). Goal: Translate a short paper calculation into a Python script. Short practice: If print(total) moves before total=total+2, what gets printed for the original **input**?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

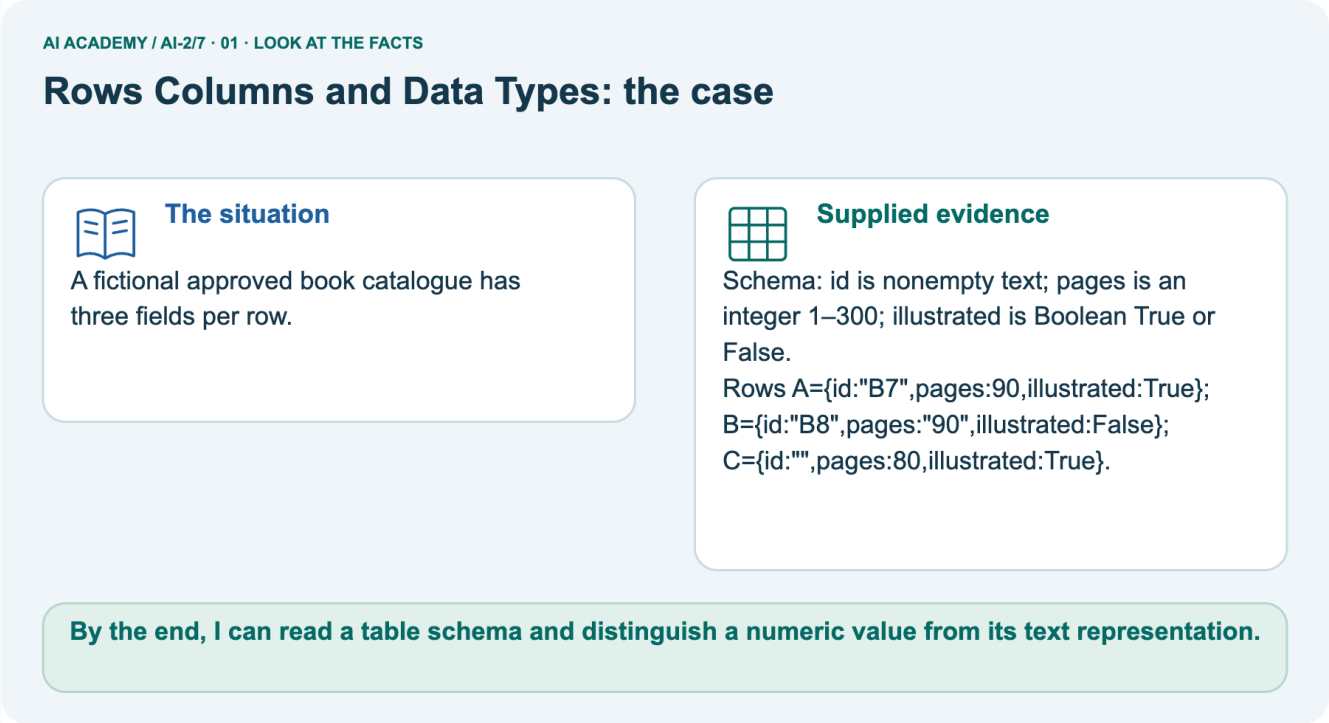
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Rows Columns and Data Types picture from its labels, then record a relevant limit.

01. Rows Columns and Data Types: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

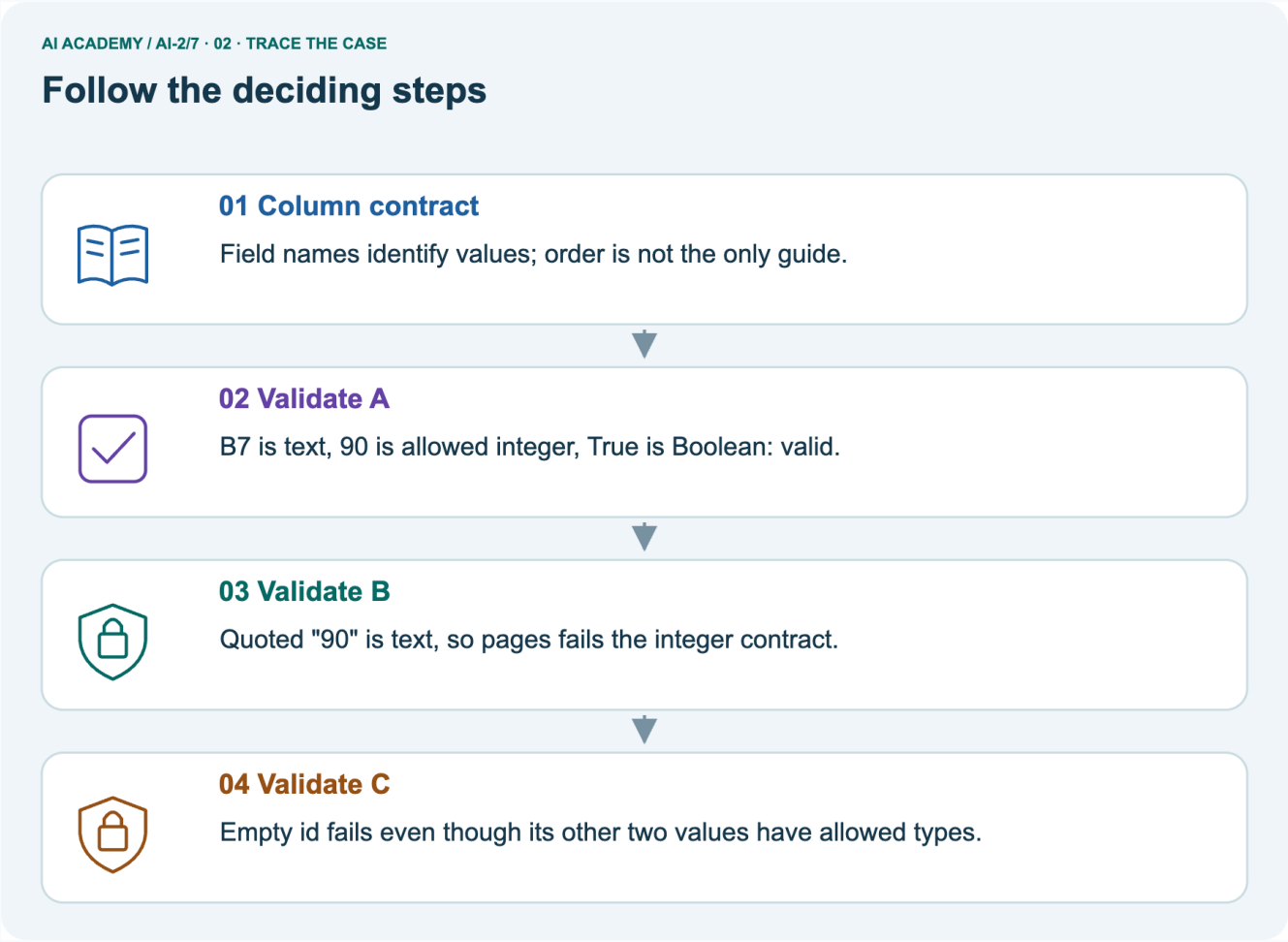
A fictional approved book catalogue has three fields per row. Supplied facts: **Schema**: id is nonempty text; pages is an integer 1–300; illustrated is Boolean True or False.

Rows A={id:"B7",pages:90,illustrated:True}; B={id:"B8",pages:"90",illustrated:False}; C={id:"",pages:80,illustrated:True}.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



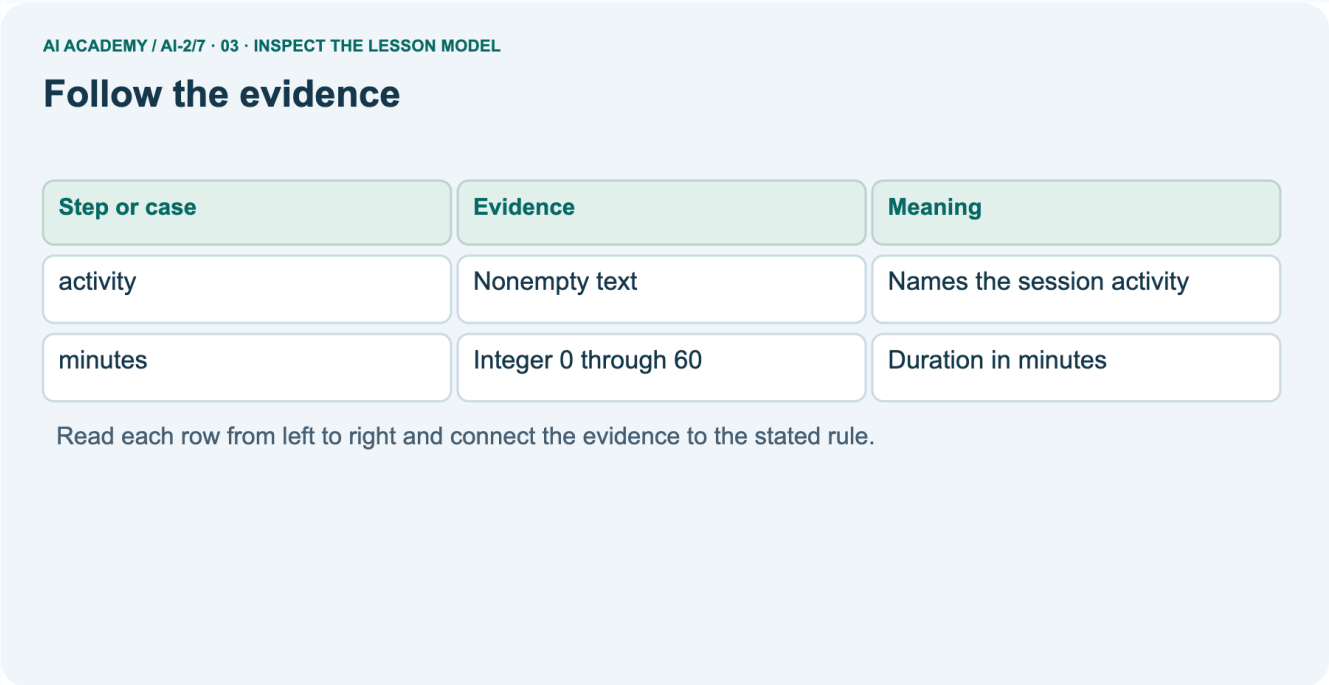
Follow the labelled picture with the matching explanation. Open the vector diagram

Column contract: Field names identify values; order is not the only guide. → Validate A: B7 is text, 90 is allowed integer, True is Boolean: valid. → Validate B: Quoted "90" is text, so pages fails the integer contract. → Validate C: Empty id fails even though its other two values have allowed **type**s.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow the evidence



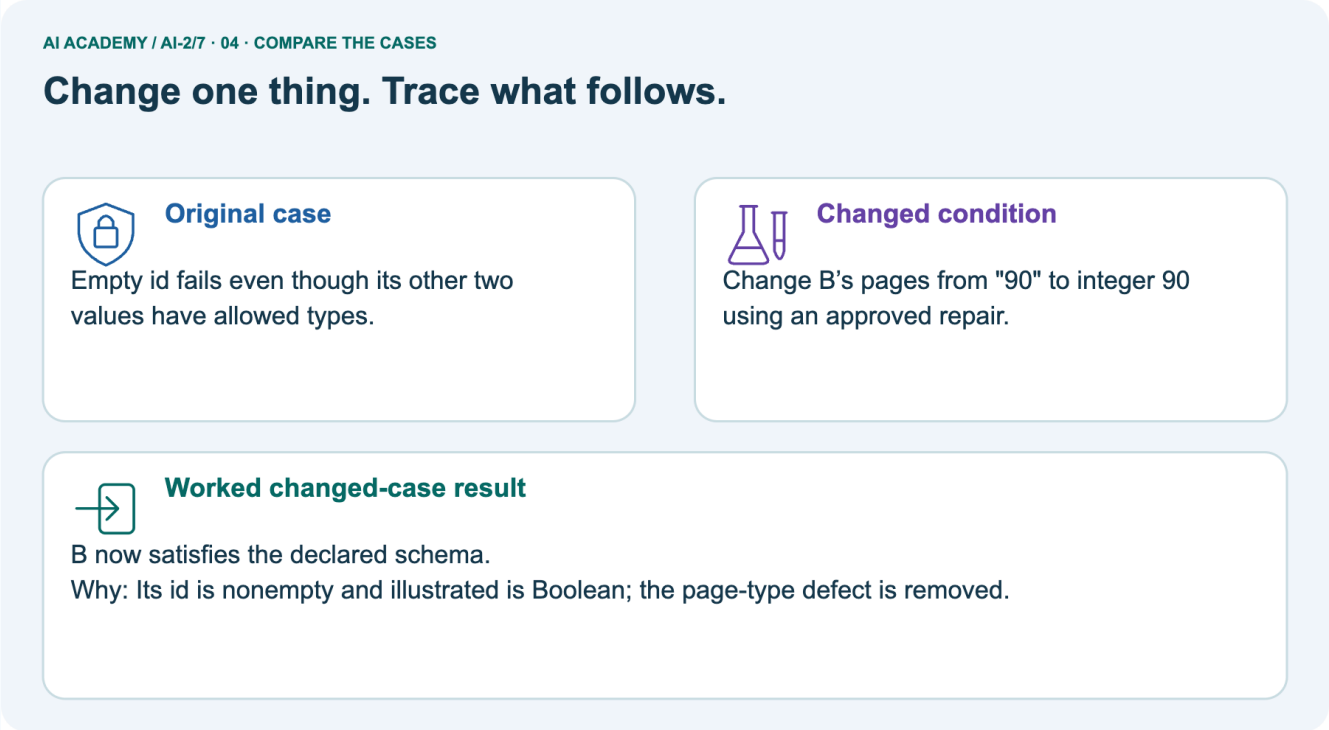
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. activity | Nonempty text | Names the session activity. minutes | Integer 0 through 60 | Duration in minutes

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

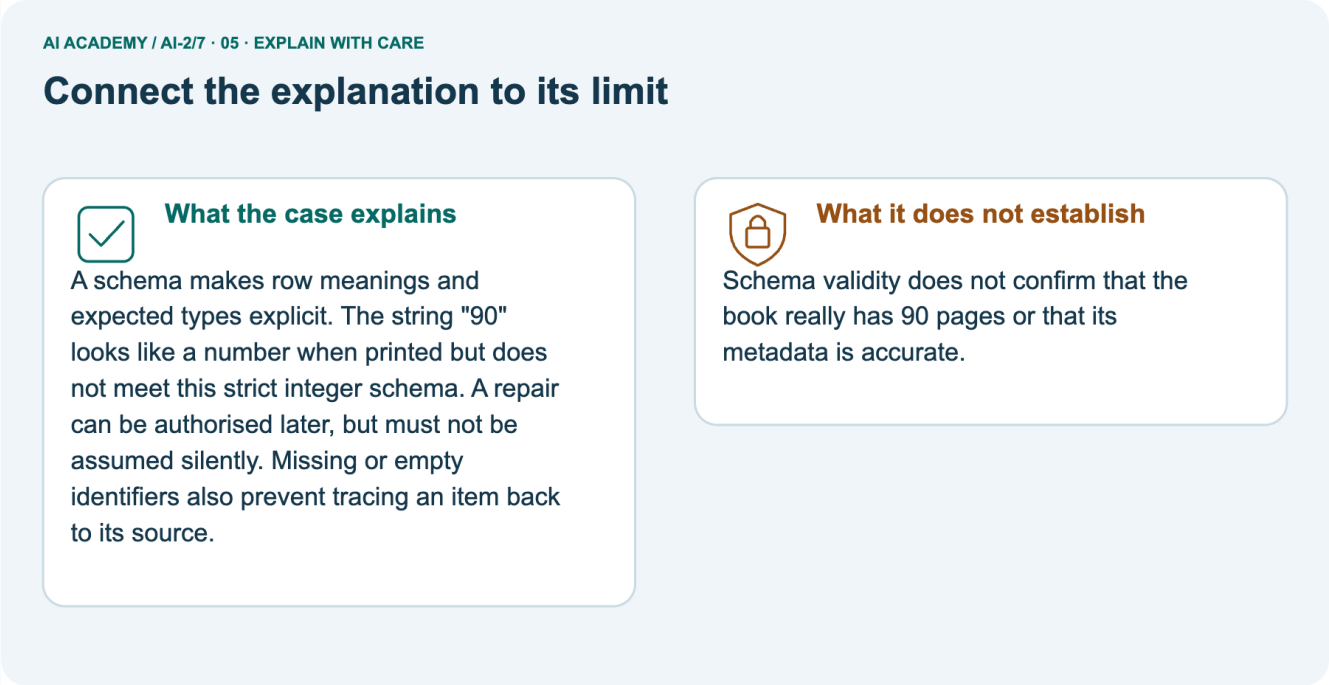
Original case: Empty id fails even though its other two values have allowed **type**s. Changed condition: Change B's pages from "90" to integer 90 using an approved repair. Expected result: B now satisfies the declared **schema**.

Reason: Its id is nonempty and illustrated is Boolean; the page-**type** defect is removed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

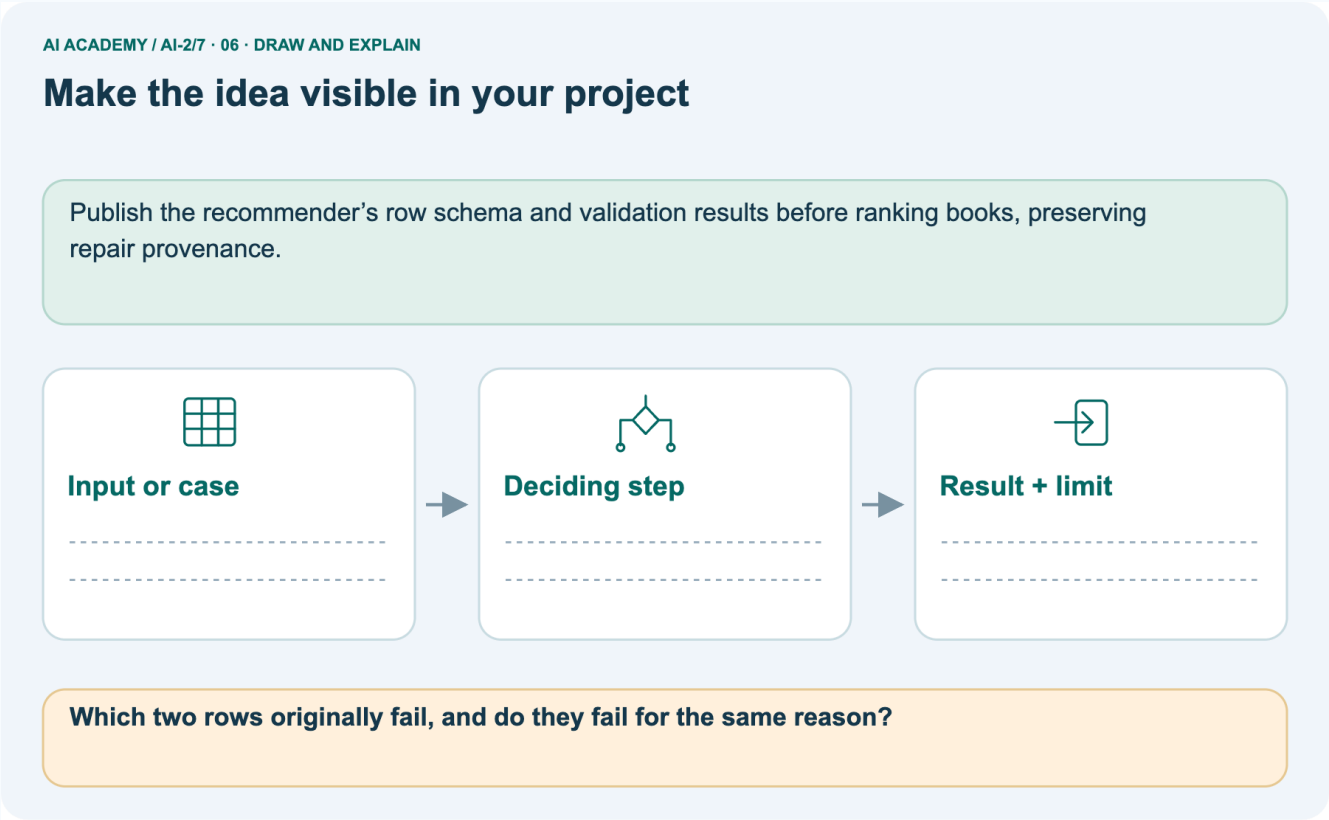
A **schema** makes row meanings and expected **type**s explicit. The string "90" looks like a number when printed but does not meet this strict integer **schema**. A repair can be authorised later, but must not be assumed silently.

Missing or empty identifiers also prevent tracing an item back to its source. Limit: **Schema** validity does not confirm that the book really has 90 pages or that its metadata is accurate.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Publish the recommender's row **schema** and validation results before ranking books, preserving repair provenance. Draw your **input**, deciding step, result and limit.

Supported question: Which two rows originally fail, and do they fail for the same reason?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.